

Redes Neurais e Aprendizado Profundo

Artur Jordão

Escola Politécnica – Engenharia de Computação e Sistemas Digitais
Universidade de São Paulo

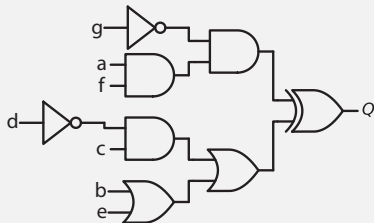
Multimodal Models

Introduction

Multimodal Models

- Multimodal models are capable of processing and understanding information from multiple modalities such as text, images, and videos
- These models play a role in solving the next generation of cognitive problems and toward a universal interface for a general-purpose assistant

Question (image and text)



Which of the alternatives gives the desired output True in this logic circuit?

- A) $a=1, b=0, c=0, d=0, e=1, f=1, g=0$
- B) $a=1, b=0, c=0, d=1, e=0, f=1, g=1$
- C) $a=1, b=0, c=0, d=1, e=0, f=0, g=1$
- D) $a=0, b=1, c=1, d=0, e=1, f=0, g=1$

Answer: **D** Qwen-3 VL 8B: **A e D**¹

GPT-5.3: **A e D** Gemini 3 Pro: **D**

¹The model becomes trapped in an endless correction loop.

Contrastive Language-Image Pre-Training (CLIP)

Introduction

Contrastive Language-Image Pre-Training (CLIP)

Learning Transferable Visual Models From Natural Language Supervision

Alec Radford^{*1} Jong Wook Kim^{*1} Chris Hallacy¹ Aditya Ramesh¹ Gabriel Goh¹ Sandhini Agarwal¹
 Girish Sastry¹ Amanda Askell¹ Pamela Mishkin¹ Jack Clark¹ Gretchen Krueger¹ Ilya Sutskever¹

Abstract

SOTA computer vision systems are trained to predict a fixed set of predetermined object categories. This restricted form of supervision limits their generality and usability since additional labeled data is needed to specify any other visual concept. Learning directly from raw text about images is a promising alternative which leverages a much broader source of supervision. We demonstrate that the simple pre-training task of predicting which caption goes with which image is an efficient and scalable way to learn SOTA image representations from scratch on a dataset of 400 million (image, text) pairs collected from the internet. After pre-training, natural language is used to reference learned visual concepts (or describe new ones) enabling zero-shot transfer of the model to downstream tasks. We study performance on over 30 different computer vision datasets, spanning tasks such as OCR, action recognition in videos, geo-localization, and many types of fine-grained object classification. The model transfers non-trivially to most tasks and is often competitive with a fully supervised baseline without the need for any dataset specific training. For instance, we

interface (McCann et al., 2018; Radford et al., 2019; Raffel et al., 2019) has enabled task-agnostic architectures to zero-shot transfer to downstream datasets. Flagship systems like GPT-3 (Brown et al., 2020) are now competitive across many tasks with bespoke models while requiring little to no dataset specific training data.

These results suggest that the aggregate supervision accessible to modern pre-training methods within web-scale collections of text surpasses that of high-quality crowd-labeled NLP datasets. However, in other fields such as computer vision it is still standard practice to pre-train models on crowd-labeled datasets such as ImageNet (Deng et al., 2009). Could scalable pre-training methods which learn directly from web text result in a similar breakthrough in computer vision? Prior work is encouraging.

Joulin et al. (2016) demonstrated that CNNs trained to predict words in image captions can learn representations competitive with ImageNet training. Li et al. (2017) then extended this approach to predicting phrase n-grams in addition to individual words and demonstrated the ability of their system to zero-shot transfer to other image classification datasets. Adopting more recent architectures and pre-training approaches, VirTex (Desai & Johnson, 2020), ICMLM (Bulent Sariyildiz et al., 2020), and ConViT (Zhong et al., 2020) have recently demonstrated the

- Modern computer vision models predict a fixed set of predefined categories
- This form of supervision limits generality and usability by requiring additional labeled data for new concepts (categories)
- Learning directly from raw text about images is a promising alternative, leveraging a much broader source of supervision

Training Overview

Contrastive Language-Image Pre-Training (CLIP)

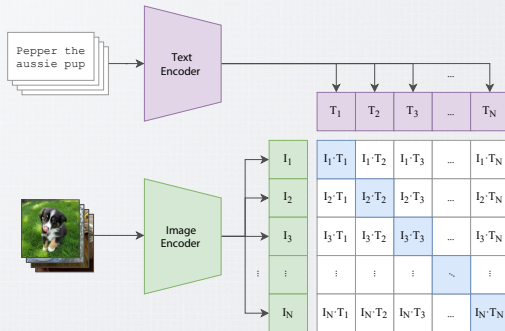
- Given N (image, text) pairs, Radford et al.¹ propose learning a model (CLIP) that predicts which of the $N \times N$ possible image–text pairings across the batch occur
- They create a new dataset of 400 million (image, text) pairs collected from a variety of publicly available sources on the Internet

¹Radford et al. *Learning Transferable Visual Models From Natural Language Supervision*. ICML, 2021

Pre-Training Phase

Contrastive Language-Image Pre-Training (CLIP)

- CLIP learns a multi-modal embedding space by jointly training an image encoder and text encoder
 - Maximize the cosine similarity of the image and text embeddings of the N **real pairs**
 - Minimize the cosine similarity of the embeddings of the $N^2 - N$ **incorrect pairs**



Zero-Shot Classification

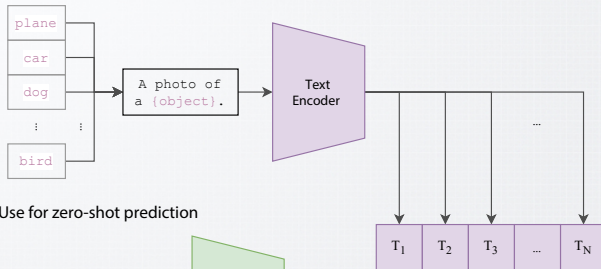
Contrastive Language-Image Pre-Training (CLIP)

- After training, CLIP exhibits strong zero-shot capabilities. For example, to classify unseen categories, CLIP requires neither retraining nor labeled examples
- Classification process
 - Encode the input image using the image encoder to obtain an image embedding
 - Encode each class description (e.g., “a photo of a bear”, “a photo of a cat”) using the text encoder to obtain text embeddings
 - Compute the similarity (e.g., cosine similarity) between the image embedding (I_1) and each text embedding (T_i)
 - **Prediction:** assign the image to the class whose text embedding achieves the highest similarity with the image embedding. Formally, $\hat{y} = \arg \max_c \text{sim}(I_1, T_i)$

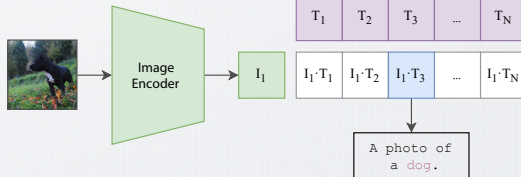
Zero-Shot Classification

Contrastive Language-Image Pre-Training (CLIP)

Create dataset classifier from label text



Use for zero-shot prediction



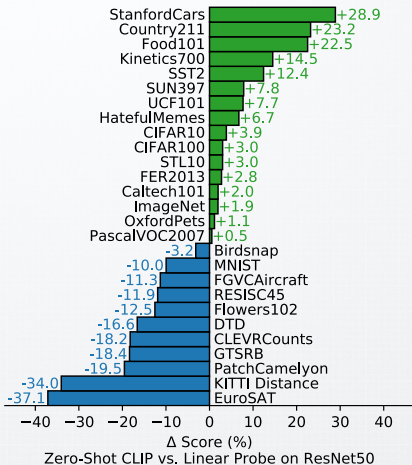
Hands-On



¹Radford et al. *Learning Transferable Visual Models From Natural Language Supervision*. ICML, 2021

Overview of Results

Contrastive Language-Image Pre-Training (CLIP)



- Linear probe: Fitting a fully supervised, regularized, logistic regression classifier on the features of the canonical ResNet50
- Zero-shot CLIP outperforms this baseline (linear probe) slightly and wins on 16 of the 27 datasets
- Zero-shot CLIP is quite weak on several specialized, complex, or abstract tasks
 - These results highlight the poor capability of zero-shot CLIP on more complex tasks

Large Language and Vision Assistant (LLaVA) Model

Introduction

Large Language and Vision Assistant (LLaVA)

Visual Instruction Tuning

Haotian Liu^{1*}, Chunyuan Li^{2*}, Qingyang Wu³, Yong Jae Lee¹

¹University of Wisconsin–Madison ²Microsoft Research ³Columbia University

<https://llava-v1.github.io>

Abstract

Instruction tuning large language models (LLMs) using machine-generated instruction-following data has been shown to improve zero-shot capabilities on new tasks, but the idea is less explored in the multimodal field. We present the first attempt to use language-only GPT-4 to generate multimodal language-image instruction-following data. By instruction tuning on such generated data, we introduce LLaVA: Large Language and Vision Assistant, an end-to-end trained large multimodal model that connects a vision encoder and an LLM for general-purpose visual and language understanding. To facilitate future research on visual instruction following, we construct two evaluation benchmarks with diverse and challenging application-oriented tasks. Our experiments show that LLaVA demonstrates impressive multimodal chat abilities, sometimes exhibiting the behaviors of multimodal GPT-4 on unseen images/instructions, and yields a 85.1% relative score compared with GPT-4 on a synthetic multimodal instruction-following dataset. When fine-tuned on Science QA, the synergy of LLaVA and GPT-4 achieves a new state-of-the-art accuracy of 92.53%. We make GPT-4 generated visual instruction tuning data, our model, and code publicly available.

- Liu et al.¹ introduce LLaVA: an effective visual instruction tuning method to turn Large Language Models (LLMs) into multi-modal models
- The work is a pioneer in the multimodal instruction tuning field

¹Liu et al. *Visual Instruction Tuning*. NeurIPS, 2023 (Oral)

Preliminaries

Large Language and Vision Assistant (LLaVA)

- Instruction-following task
 - Interactivity and adaptability to user instructions
- Instruction tuning
 - Focuses on improving the model instruction-following capacity



User: Is this unusual? Please explain in detail.



User: What's happening in the desert?



User: Please read the text in this image and return the information in the following JSON format

¹Liu et al. *Visual Instruction Tuning*. NeurIPS, 2023 (Oral)

Instruction-following Data

Large Language and Vision Assistant (LLaVA)

- There is a lack of vision-language instruction-following data
 - Liu et al.¹ present the first attempt to use language-only GPT-4 to generate multimodal language-image instruction-following data
- For an image $\mathbf{X}_v$ and its associated caption $\mathbf{X}_c$, it is natural to create a set of questions $\mathbf{X}_q$ with the intent to instruct the assistant to describe the image content
 - Liu et al.¹ prompt GPT-4 to curate such a list of questions
 - A simple way to expand an image-text pair to its instruction-following version is
Human : $\mathbf{X}_q$ $\mathbf{X}_v$ <STOP> Assistant : $\mathbf{X}_c$ <STOP>
 - Though cheap to construct, this simple expanded version lacks diversity and in-depth reasoning in both the instructions and responses

¹Liu et al. *Visual Instruction Tuning*. NeurIPS, 2023 (Oral)

Instruction-following Data

Large Language and Vision Assistant (LLaVA)

- In order to encode an image into its visual features to prompt a text-only GPT, Liu et al.¹ use two types of symbolic representations
 - Captions. Describe the visual scene from various perspectives
 - Bounding boxes. Localize the objects in the scene, and each box encodes the object concept and its spatial location

Context type 1: Captions

A group of people standing outside of a black vehicle with various luggage.

Luggage surrounds a vehicle in an underground parking area

People try to fit all of their luggage in an SUV.

The sport utility vehicle is parked in the public garage, being packed for a trip

Some people with luggage near a van that is transporting it.

Context type 2: Boxes

person: [0.681, 0.242, 0.774, 0.694], backpack: [0.384, 0.696, 0.485, 0.914], suitcase: ...



¹Liu et al. *Visual Instruction Tuning*. NeurIPS, 2023 (Oral)

Instruction-following Data

Large Language and Vision Assistant (LLaVA)

- Types of instruction-following data
 - Conversation. Conversation between the assistant and a person asking questions about an image. Here, answers are in a tone as if the assistant is seeing the image and answering the question
 - Detailed description. Questions that include a rich and comprehensive description for an image
 - Complex reasoning. In-depth reasoning questions where the answers often require a step-by-step reasoning process by following rigorous logic

Instruction-following data

Large Language and Vision Assistant (LLaVA)

Context type 1: Captions

A group of people standing outside of a black vehicle with various luggage.

Luggage surrounds a vehicle in an underground parking area

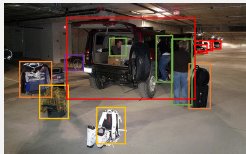
People try to fit all of their luggage in an SUV.

The sport utility vehicle is parked in the public garage, being packed for a trip

Some people with luggage near a van that is transporting it.

Context type 2: Boxes

person: [0.681, 0.242, 0.774, 0.694], backpack: [0.384, 0.696, 0.485, 0.914], suitcase: ...



Response type 1: conversation

Question: What type of vehicle is featured in the image?

Answer: The image features a black sport utility vehicle (SUV) ...

Response type 2: detailed description

The image is an underground parking area with a black sport utility vehicle (SUV) parked. There are three people in the scene, with one person standing closer to the left side of the vehicle, another person in the middle, and the third person on the right side. They are all working together to pack their luggage into the SUV for a trip. ...

Response type 3: complex reasoning

Question: What challenges do these people face?

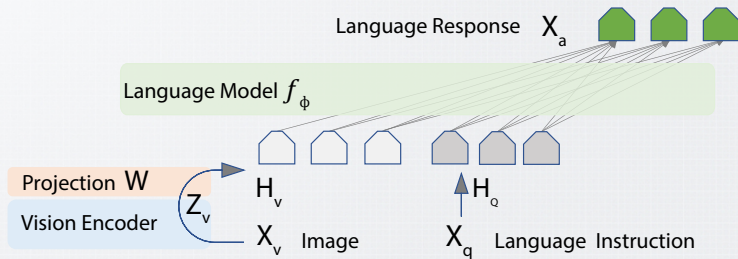
Answer: In the image, a group of people is standing outside a black SUV in a parking area, surrounded by various pieces of luggage, including suitcases and backpacks. They are facing the challenge of fitting all their luggage into the black SUV. There are multiple suitcases and backpacks to be packed, which suggests that the group has a significant amount of belongings ...

¹Liu et al. *Visual Instruction Tuning*. NeurIPS, 2023 (Oral)

Model Architecture

Large Language and Vision Assistant (LLaVA)

- LLaVA leverages the capabilities of both the pre-trained LLM and visual model
 - LLM: Vicuna
 - Visual encoder: CLIP
 - The architecture employs a simple linear layer (projection matrix W) to connect image features (Z_v) into the word embedding (H_v) space



Training

Large Language and Vision Assistant (LLaVA)

- Stage 1. Pre-training for Feature Alignment
 - Update the projection matrix (W) only
 - The visual encoder and LLM weights frozen
- Stage 2. Fine-tuning End-to-End
 - Update both the pre-trained weights of the projection layer and the LLM
 - The trainable parameters are $\theta = \{W, \phi\}$
- Visual encoder weights always frozen

¹Liu et al. *Visual Instruction Tuning*. NeurIPS, 2023 (Oral)

Overview of the Results

Large Language and Vision Assistant (LLaVA)

- Liu et al.¹ employ GPT-4 (**LLM as a Judge**) to measure the quality of generated responses
 - The judge evaluates the helpfulness, relevance, accuracy, and level of detail of the responses
 - It gives an overall score on a scale of 1 to 10 (the higher the better)

Challenging examples from LLaVA-Bench (In-the-Wild)



Annotation. A close-up photo of a meal at ICHIRAN. The [...]

Question 1. What's the name of the restaurant?

Question 2. Describe this photo in detail.

¹Liu et al. *Visual Instruction Tuning*. NeurIPS, 2023 (Oral)

Overview of the Results

Large Language and Vision Assistant (LLaVA)

- LLaVA-Bench (In-the-Wild)
 - Set of 24 images with 60 questions, including indoor and outdoor scenes, memes, paintings, sketches

	Conversation	Detail description	Complex reasoning	All
OpenFlamingo ²	19.3 ± 0.5	19.0 ± 0.5	19.1 ± 0.7	19.1 ± 0.4
BLIP-2 ³	54.6 ± 1.4	29.1 ± 1.2	32.9 ± 0.7	38.1 ± 1.0
LLaVA ¹	57.3 ± 1.9	52.5 ± 6.3	81.7 ± 1.8	67.3 ± 2.0

¹Liu et al. *Visual Instruction Tuning*. NeurIPS, 2023 (Oral)

²Awadalla et al. *Openflamingo*. 2023

³Li et al. *BLIP-2: Bootstrapping Language-Image Pre-training with Frozen Image Encoders and Large Language Models*. ICML, 2023

LLaVa Variants and Monolithic Models

Introduction

LLaVa Variants and Monolithic Models



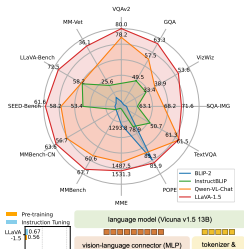
This CVPR paper is the Open Access version, provided by the Computer Vision Foundation.
Except for this watermark, it is identical to the accepted version;
the final published version of the proceedings is available on IEEE Xplore.

Improved Baselines with Visual Instruction Tuning

Haotian Liu¹ Chunyuan Li² Yuheng Li¹ Yong Jae Lee¹
¹University of Wisconsin–Madison ²Microsoft Research, Redmond
<https://llava-vl.github.io>

Abstract

Large multimodal models (LMM) have recently shown encouraging progress with visual instruction tuning. In this paper, we present the first systematic study to investigate the design choices of LMMs in a controlled setting under the LLaVA framework. We show that the fully-connected vision-language connector in LLaVA is surprisingly powerful and data-efficient. With simple modifications to LLaVA, namely, using CLIP-ViT-L-336px with an MLP projection and adding academic-task-oriented VQA data with response formatting prompts, we establish stronger baselines that achieve state-of-the-art across 11 benchmarks. Our final 13B checkpoint uses merely 1.2M publicly available data, and finishes full training in ~ 1 day on a single 8-A100 node. Furthermore, we present some early exploration of open problems in LMMs, including scaling to higher resolution inputs, compositional capabilities, and model hallucination, etc. We hope this makes state-of-the-art LMM research more



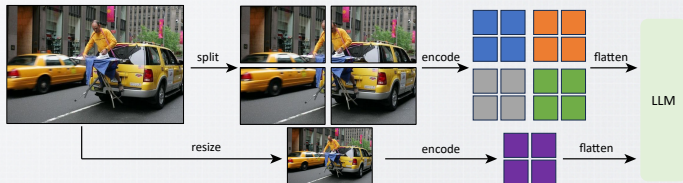
- Since introduction of LLaVa¹, most multimodal models generally follow a common approach of visual encoder → cross-modal connector → LLM
- The LLaVA architecture by Liu et al.¹ limits the resolution of the image to a fixed size that the LMM accepts during inference
- To overcome this issue, Liu et al.² proposed to divide the image into smaller image patches and encode them independently

¹Liu et al. *Visual Instruction Tuning*. NeurIPS, 2023 (Oral)

LLaVa Variants

LLaVa Variants and Monolithic Models

- Liu et al.¹ propose to obtain feature maps of individual patches (smaller image patches)
- After obtaining the feature maps, the model combines them into a single large feature map of the target resolution, and feed that into the LLM



¹Liu et al. *Improved Baselines with Visual Instruction Tuning*. CVPR, 2024

Qwen

LLaVa Variants and Monolithic Models

- Wang et al.¹ propose to enhance the model's adaptability to varying resolutions
 - No single fixed resolution achieves optimal performance across all benchmarks
 - Merely increasing the image size does not always lead to improved performance
- Three-stage training
 - Train only the Vision Transformer (ViT)
 - Unfreeze all parameters and train on broader data
 - Freeze the ViT again and fine-tune only the LLM using instruction datasets



Hands-On

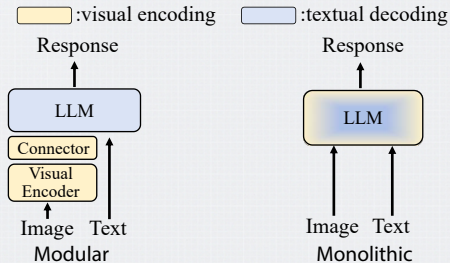


¹Wang et al. *Qwen2-VL: Enhancing Vision-Language Model's Perception of the World at Any Resolution.*

Monolithic Models

LLaVa Variants and Monolithic Models

- Multimodal Models adopt a modular architecture
 - The architecture processes visual encoding and language decoding separately
 - Requires an intermediate model to project visual features into the LLM embedding space
- Recent research explores monolithic models
 - Integrate visual perception and multimodal understanding directly into a **single** LLM
 - Visual Encoder-free (lightweight structure)



Monolithic Models

LLaVa Variants and Monolithic Models

- Modular models achieve superior performance
- Monolithic models show better potential in terms of design simplicity and deployment efficiency

Model	#A-Param	AvgMM	AvgQA
▼ <i>Modular MLLMs:</i>			
MobileVLM-V2-3B	3.0B	—	—
Mini-Gemini-2B	3.5B	—	—
MM1-3B-MoE-Chat	3.5B	—	—
DeepSeek-VL-1.3B	2.0B	43.4	—
PaliGemma-3B	2.9B	46.7	—
MiniCPM-V-2	2.8B	51.2	—
† InternVL-1.5-2B	2.2B	54.4	71.7
Qwen2VL-2B	2.1B	—	—
▼ <i>Monolithic MLLMs:</i>			
Fuyu-8B (HD)	8B	—	—
SOLO	7B	—	—
Chameleon-7B	7B	16.1	17.9
EVE-7B	7B	34.8	40.8
EVE-7B (HD)	7B	38.9	54.6
Emu3	8B	—	67.6
Mono-InternVL-2B	1.8B	55.2	70.1

Luo et al. *Mono-InternVL: Pushing the Boundaries of Monolithic Multimodal Large Language Models with Endogenous Visual Pre-training*. CVPR, 2025

LLM as a Judge

Introduction

LLM as a Judge

Judging LLM-as-a-Judge with MT-Bench and Chatbot Arena

Lianmin Zheng^{1*} Wei-Lin Chiang^{1*} Ying Sheng^{4*} Siyuan Zhuang¹

Zhanghao Wu¹ Yonghao Zhuang³ Zi Lin² Zhuohan Li¹ Dacheng Li^{1,3}

Eric P. Xing^{1,5} Hao Zhang^{1,2} Joseph E. Gonzalez¹ Ion Stoica¹

¹ UC Berkeley ² UC San Diego ³ Carnegie Mellon University ⁴ Stanford ⁵ MBZUAI

Abstract

Evaluating large language model (LLM) based chat assistants is challenging due to their broad capabilities and the inadequacy of existing benchmarks in measuring human preferences. To address this, we explore using strong LLMs as judges to evaluate these models on more open-ended questions. We examine the usage and limitations of LLM-as-a-judge, including position, verbosity, and self-enhancement biases, as well as limited reasoning ability, and propose solutions to mitigate some of them. We then verify the agreement between LLM judges and human preferences by introducing two benchmarks: MT-bench, a multi-turn question set; and Chatbot Arena, a crowdsourced battle platform. Our results reveal that strong LLM judges like GPT-4 can match both controlled and crowdsourced human preferences well, achieving over 80% agreement, the same level of agreement between humans. Hence, LLM-as-a-judge is a scalable and explainable way to approximate human preferences, which are otherwise very expensive to obtain. Additionally, we show our benchmark and traditional benchmarks complement each other by evaluating several variants of LLaMA and Vicuna. The MT-bench questions, 3K expert votes, and 30K conversations with human preferences are publicly available at https://github.com/lm-sys/FastChat/tree/main/fastchat/llm_judge.

- Evaluating LLMs is challenging due to their broad capabilities and the inadequacy of existing benchmarks in **measuring human preferences**
- Strong LLM judges (i.e., GPT-4) can match both controlled and crowdsourced human preferences well
- LLM-as-a-judge is a scalable and explainable way to approximate human preferences

Types of LLM-as-a-Judge

LLM as a Judge

- Pairwise comparison
 - An LLM judge receives a question and two answers and determines which answer is better or declares a tie

Default prompt for pairwise comparison

[System] Please act as an impartial judge and evaluate the quality of the responses provided by two AI assistants to the user question displayed below. [...] After providing your explanation, output your final verdict by strictly following this format: "[[A]]" if assistant A is better, "[[B]]" if assistant B is better, and "[[C]]" for a tie.

[User Question]

{question}

[The Start of Assistant A's Answer]

{answer_a}

[The End of Assistant A's Answer]

[The Start of Assistant B's Answer]

{answer_b}

[The End of Assistant B's Answer]

Types of LLM-as-a-Judge

LLM as a Judge

- Single answer grading
 - An LLM judge directly assigns a score to a single answer

Default prompt for single answer grading

[System] Please act as an impartial judge and evaluate the quality of the response provided by an AI assistant to the user question displayed below. [...] After providing your explanation, please rate the response on a scale of 1 to 10 by strictly following this format: "[[rating]]", for example: "Rating: [[5]]".

[Question]

{question}

[The Start of Assistant's Answer]

{answer}

[The End of Assistant's Answer]

Types of LLM-as-a-Judge

LLM as a Judge

- Reference-guided grading
 - In certain cases, it may be beneficial to provide a reference solution if applicable

Prompt for reference-guided pairwise comparison

[System] [...] After providing your explanation, output your final verdict by strictly following this format: "[[A]]" if assistant A is better, "[[B]]" if assistant B is better, and "[[C]]" for a tie.

[User Question]

{question}

[The Start of Reference Answer]

{answer_ref}

[The End of Reference Answer]

[The Start of Assistant A's Answer]

{answer_a}

[The End of Assistant A's Answer]

[The Start of Assistant B's Answer]

{answer_b}

[The End of Assistant B's Answer]

Limitations of LLM-as-a-Judge

LLM as a Judge

- LLM-as-a-judge offers two key benefits: *scalability* and *explainability*
 - It reduces the need for human involvement, enabling scalable benchmarks and fast iterations
- Zheng et al.¹ identify certain biases and limitations of LLM judges
 - Position bias
 - Verbosity bias
 - Self-enhancement bias
 - Limited capability in grading math and reasoning questions

¹Zheng et al. *Judging LLM-as-a-Judge with MT-Bench and Chatbot Arena*. NeurIPS, 2023

Limitations of LLM-as-a-Judge

LLM as a Judge

- Position bias
 - An LLM exhibits a propensity to favor certain positions over others

Prompt for reference-guided pairwise comparison

Question: What are some business etiquette norms when doing business in Japan?

Assistant A:

Here are some business etiquette norms to keep in mind when doing business in Japan:

1. Bowing: Bowing is a [...]

Assistant B:

When doing business in Japan, it is important to be aware of and respect the country's business [...]

1. Greetings: A handshake and [...]

GPT-4 Judgment (when A is placed in the first position):

[...] However, Assistant A's answer is more detailed and organized [...]

GPT-4 Judgment (when B is placed in the first position):

[...] However, Assistant B's answer is more detailed and covers a wider range of topics [...]

Limitations of LLM-as-a-Judge

LLM as a Judge

- Verbosity bias
 - An LLM judge favors longer, verbose responses, even if they are not as clear, high-quality, or accurate as shorter alternatives
- Self-enhancement bias
 - LLM judges may favor the answers generated by themselves
- Limited capability in grading math and reasoning questions
 - An LLM shows limitations in grading basic math problems **which it is capable of solving**